Microelectronics

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1 Microelectronics

2 Signals, Amplifiers, and Operational Amplifiers

I Signal

i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems on voltage or current signals, especially the former. Only voltage and current signals are studied here.

ii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

iii Cascade

Output of a component as the input of another.

iv Transducer

A transducer is a device that convert a signal to a voltage or current signal or vice versa.

v Signal processing

Signal processing includes analog signal processing (ASP), analog-to-digital converter (ADC), digital signal processor (DSP), digital-to-analog converter (DAC), amplifier, etc.

vi Coupling capacitor

A capacitor connected in series between stages to pass AC signal while blocking DC.

vii Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g. $i_C(t)$.
- Lowercase character with lowercase subscript means AC signal, e.g. $i_c(t)$.
- Uppercase character with uppercase subscript means DC signal, e.g. I_C.
- Uppercase character with lowercase subscript means peak signal, e.g. I_c.

Their relation is

$$i_C(t) = i_c(t) + I_C$$

$$i_c(t) = I_c \sin(\omega t + \varphi)$$

viii Differential and common-mode signal

For two input signals, the differential signal (AC signal) is the difference between them, and the common-mode signal (DC signal) is the algebraic average of them.

ix Input and output impedance

For a component's input, input impedance is the impedance relative to ground as seemed from the input.

For a component's output, output impedance is the impedance relative to ground as seemed from the input.

x Differential and common-mode impedance

For two input terminals, the differential input impedance is the input impedance relative to ground as seemed by them when a ideal source connects them directly.

For two input terminals, the common-mode input impedance is the input impedance relative to ground as seemed by them when an ideal source's output connects both of them.

xi Differential and common-mode transfer function

For two input terminals, the differential transfer function is the transfer function when they are connected to $\frac{1}{2}$ differential input signal and $-\frac{1}{2}$ differential input signal respectively.

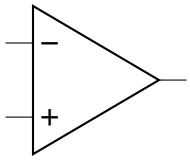
For two input terminals, the common-mode transfer function is the transfer function when an ideal source's output connecting both of them is treated as input signal.

II Amplifier (amp)

i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain A and optionally convert voltage signal to current signal or vice versa.

ii Diagram



iii Two-port model

An amplifier can be modeled as a two port model, where an one input port and one output port is related by the amplifier, which acts as a controlled source controlled by the input port voltage or current which enforces voltage or current on the output port.

iv Circuit model of voltage amplifier

A voltage amplifier draws current from a source v_s with internal resistance R_s with input resistance R_i and voltage v_{in} across it and supplies $v_c = \mu v_{in}$ to a load network with output resistance R_o and load R_L, where μ, also denoted A_{vo}, is open-loop voltage gain factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{v_c}{v_{in}} = \mu$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu$$

v Circuit model of current amplifier

A current amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $i_c = \beta i_{in}$ by CCCS to a load network with output resistance R_o in parallel and load R_L, where β, also denoted A_{cs}, is short-circuit current gain factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{i_c}{i_{in}} = \beta$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta$$

vi Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source v_s with internal resistance R_s in series with input resistance R_i and current i_{in} through it and supplies $i_c = g v_{in}$ by VCCS to a load network with output resistance R_o in parallel and load R_L, where μ is open-loop transconductance factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{i_c}{v_{in}} = g$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g\frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g$$

vii Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $v_c = ri_{in}$ by CCVS to a load network with output resistance R_o in series and load R_L, where r is short-circuit transresistance factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{v_c}{i_{in}} = r$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = \beta\frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r$$

viii Amplifier gain in decibels (dB)

A power gain A_p in dB is defined as

$$G_p = 10\log_{10}(A_p)$$

A voltage gain A_v in dB is defined as

$$G_v = 20\log_{10}(A_v)$$

A current gain A_c in dB is defined as

$$G_c = 20\log_{10}(A_c)$$

ix Saturation

The linear characteristic only remains within a range. A positive and negative saturation levels can be observed.

x S parameter

If the energy enters through port i and emerges from port j, the S parameter is named S_{ji}.

xi Unilateral amplifier

An unilateral amplifier is an amplifier in which the signal transfer is effectively one-way, that is, the output signal is only dependent on the input signal but not what is connected to the output terminal. That is for each output port j and each input port i, $S_{ij} = 0$.

xii Power supply

To amplify signal, power supplies are needed, but often omitted in the diagram.

- V_{CC+} or V_{CC}: positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the − side of an op-amp or omitted.

The DC power delivered is the sum of power from the positive and negative power supply terminals V_{CC} and V_{EE}

$$P_{DC} = V_{CC} I_{CC} + V_{EE} I_{EE}$$

Suppose input and output signals have power P_I and P_L.

$$P_{DC} + P_I = P_L + P_{\text{dissipated}}$$

where $P_{\text{dissipated}}$ is the power dissipated.

xiii Classification based on frequency response

A band-pass amplifier is also called tuned amplifier. A low-pass amplifier is also called DC amplifier. A amplifier directly coupled with a coupling capacitor, called capacitively coupled, is called AC amplifier.

xiv Output impedance

In Laplace domain, for an amplifier with gain $A = \frac{V_o}{V_i}$ and output impedance Z_o driving a load impedance Z_L, the output voltage appears across the load V_L is:

$$V_L = V_o \frac{Z_L}{Z_o + Z_L},$$

and the gain as seemed from the load $A_L = \frac{V_L}{V_i}$ is:

$$A_L = A \frac{Z_L}{Z_o + Z_L},$$

and the power output to the load is

$$P_L = \frac{{V_L}^2}{Z_L}$$

xv Input impedance

In Laplace domain, for an amplifier with input impedance Z_i, the power input is:

$$P_i = \frac{{V_i}^2}{Z_i}$$

xvi Power gain

For an amplifier with gain A, input impedance Z_i, and output impedance Z_o driving a load impedance Z_L, the power gain (Laplace domain) is:

$$G_P = \frac{P_L}{P_i} = {A_L}^2 \frac{Z_i}{Z_L} = \frac{Z_i Z_L}{\left(Z_o + Z_L\right)^2}$$

III Operational Amplifier (op amp or op-amp)

i Definition

An op amp has five terminals:

- v_+: non-inverting input voltage of the terminal at the $+$ sign;
- v_-: inverting input voltage of the terminal at the $-$ sign;
- v_o: output voltage of the terminal at the opposite side of inputs;
- V_{CC+} or V_{CC}: positive supply voltage, denoted on the $+$ side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the $-$ side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os})$$

where A is (differential) open-loop gain or open-circuit gain, and v_{os} is input offset voltage.

$$v_{ld} = v_+ - v_-$$
$$v_{lcm} = \frac{v_+ + v_-}{2}$$

ii Ideal op amp

- Infinite input impedance: No signal current flowing into v_+ and v_-.
- Infinite open-loop gain.
- Infinite bandwidth: The open-loop gain remains constant in all frequencies.
- Zero output impedance: The amplifier is unilateral.
- Zero input offset voltage: $v_{os} = 0$, which implies zero common-mode gain, i.e., infinite common-mode rejection.

This is a linear element with inputs v_+ and v_- and output v_o.

v_o may be any voltage since $A = \infty$ and $0 * \infty$ can be any number in $[0, \infty]$ in limit sense.

The open-loop gain of a real op amp is finite and dependent on frequency.

iii Internally compensated op amp

An internally compensated op amp is am op amp whose open-loop transfer function $A(s)$ is STC low-pass filter to ensure the op amp is stable,

$$A(s) = \frac{A_0}{1 + \frac{s}{\omega_b}} = \frac{\omega_t}{s + \omega_b}, \quad \omega_t = A_0 \omega_b,$$

where $A_0 \gg 1$ is DC gain, ω_b is 3 dB bandwidth, and ω_t is approximately the frequency of which the gain $|A|$ reach unity (0 dB):

$$|A(j\omega)| = \frac{A_0}{\sqrt{1 + \left(\frac{\omega}{\omega_b}\right)^2}} = 1$$

$$A_0{}^2 = 1 + \left(\frac{\omega}{\omega_b}\right)^2$$

$$\omega = \sqrt{A_0{}^2 - 1}\omega_b \approx A_0 \omega_b$$

For $\omega \gg \omega_b$,

$$|A(j\omega)|\, \omega \approx A_0 \omega_b$$

The non-angular frequency $f_t = \frac{\omega_t}{2\pi}$ is usually specified on the op-amp data sheet as the unity-gain bandwidth or gain-bandwidth product (GBW, GBWP, GBP, or GB). Such op amp is said to have a single-pole model since the single pole dominates the amplifier responses, also called dominant pole.

iv Feedback configuration

To have stable signal, op amp in a circuit is connected with a negative feedback configuration, which consists of the op amp and a connection from v_o to either of the input terminals with a resistance on it, called feedback connection. When the feedback connection connects to v_-, it is called inverting configuration. When the feedback connection connects to v_+, it is called non-inverting configuration.

We can construct a virtual short circuit between v_+ and v_- since it will becomes $v_+ = v_-$ immediately due to the feedback.

v Inverting amplifier

An inverting amplifier consists of an op amp where input voltage v_i is connected to v_- through an input resistance R_i (current from v_i to v_-), v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistance R_f (current from v_- to v_o).

$$v_+ = v_- = 0$$

$$i_i = \frac{v_i}{R_i} = i_f = -\frac{v_o}{R_f}$$

$$v_o = -\frac{R_f}{R_i} v_i$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = -\frac{R_f}{R_i}$$

The feedback factor β is defined as

$$\beta = \frac{1}{1 + \frac{R_f}{R_i}}$$

Effect of finite open-loop gain:

$$v_+ = 0$$

$$v_o = -Av_-$$

$$i_i = \frac{v_i - v_-}{R_i} = \frac{v_i + \frac{v_o}{A}}{R_i} = i_f = -\frac{v_o + \frac{v_o}{A}}{R_f}$$

$$G = \frac{v_o}{v_i} = \frac{-\frac{R_f}{R_i}}{1 + \frac{1+\frac{R_f}{R_i}}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{1-G_{\text{ideal}}}{A}}$$

$$\lim_{A\to\infty} G = -\frac{R_f}{R_i} = G_{\text{ideal}}$$

For single-pole model $A(s) = \frac{A_0}{1 + \frac{s}{\omega_b}}$ with $A_0 \gg 1 + \frac{R_f}{R_i}$ and $\omega_t = A_0\omega_b$:

$$\begin{aligned} G &= \frac{-\frac{R_f}{R_i}}{1 + \frac{1+\frac{R_f}{R_i}}{A}} \\ &= \frac{-\frac{R_f}{R_i}}{1 + \frac{1+\frac{R_f}{R_i}}{A_0} + \frac{\frac{s}{\omega_t}}{1+\frac{R_f}{R_i}}} \\ &\approx \frac{-\frac{R_f}{R_i}}{1 + \frac{\frac{s}{\omega_t}}{1+\frac{R_f}{R_i}}}, \end{aligned}$$

which is a low-pass STC filter with 3 dB bandwidth $\frac{\omega_t}{1 + \frac{R_f}{R_i}} = \omega_t\beta$ and DC gain $-\frac{R_f}{R_i}$.

vi Transresistance amplifier

A transresistance amplifier consists of an op amp where input current i_i is connected to v_-, v_+ is connected to ground, and v_o is fed back to v_- through two resistances R_o (current from x to v_o) and R_f (current from v_- to x) in series connection, with the node x between R_o and R_f connected to ground through a resistance R_g (current from x to ground).

$$v_x = -R_f i_i$$

$$i_g = -\frac{R_f}{R_g} i_i$$

$$i_o = \left(1 + \frac{R_f}{R_g}\right) i_i$$

$$v_o = -R_f i_i - R_o i_o = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right) i_i$$

$$\frac{v_o}{i_i} = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right)$$

vii Weighted summing amplifier

A weighted summing amplifier consists of an op amp, input voltages $v_1, v_2, \ldots, v_n$, n input resistors $R_1, R_2, \ldots, R_n$, and a feedback resistor R_f, where for any $k \in \mathbb{N} \wedge k \leq n$, v_k is connected to v_- through R_k, v_+ is connected to ground, and v_o is fed back to v_- through R_f.

$$i_k = \frac{v_k}{R_k}$$

$$\sum_{k=1}^{n} i_k = i_f = \frac{-v_o}{R_f}$$

$$v_o = -R_f \sum_{k=1}^{n} \frac{v_k}{R_k}$$

When $R_k = R$ for all $k \in \mathbb{N} \wedge k \leq n$:

$$v_o = -\frac{R_f}{R} \sum_{k=1}^{n} v_k$$

viii Non-inverting amplifier

A non-inverting (or noninverting) amplifier consists of an op amp where input voltage v_i is connected to v_+, v_- is connected ground through a ground resistance R_g (current from v_- to ground), and v_o is fed back to v_- through a feedback resistance R_f (current from v_o to v_-).

$$v_+ = v_- = v_i$$

$$i_g = \frac{v_i}{R_g} = i_f = \frac{v_o - v_i}{R_f}$$

$$v_o = \left(1 + \frac{R_f}{R_g}\right) v_i$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_g}$$

The feedback factor β is defined as

$$\beta = \frac{1}{1 + \frac{R_f}{R_g}}$$

Effect of finite open-loop gain:

$$v_+ = v_i$$

$$v_o = A(v_i - v_-)$$

$$v_- = v_i - \frac{v_o}{A}$$

$$i_g = \frac{v_-}{R_g} = \frac{v_i - \frac{v_o}{A}}{R_g} = i_f = \frac{v_o - v_-}{R_f} = \frac{v_o - v_i + \frac{v_o}{A}}{R_f}$$

$$G = \frac{v_o}{v_i} = \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{G_{\text{ideal}}}{A}}$$

$$\lim_{A \to \infty} G = 1 + \frac{R_f}{R_g} = G_{\text{ideal}}$$

For single-pole model $A(s) = \frac{A_0}{1 + \frac{s}{\omega_b}}$ with $A_0 \gg 1 + \frac{R_f}{R_g}$ and $\omega_t = A_0 \omega_b$:

$$\begin{aligned} G &= \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A}} \\ &= \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A_0} + \frac{\frac{s}{\omega_t}}{\frac{R_f}{1 + \frac{R_f}{R_g}}}} \\ &\approx \frac{1 + \frac{R_f}{R_g}}{1 + \frac{\frac{s}{\omega_t}}{\frac{R_f}{1 + \frac{R_f}{R_g}}}} \end{aligned}$$

which is a low-pass STC filter with 3 dB bandwidth $\frac{\omega_t}{1 + \frac{R_f}{R_g}} = \omega_t \beta$ and DC gain $1 + \frac{R_f}{R_g}$.

ix Unity gain amplifier, unity gain buffer, buffer amplifier, or voltage follower

A unity gain amplifier consist of an op amp where input voltage v_i is connected to v_+, and v_o is fed back to v_-.

$$v_o = v_- = v_+ = v_i$$

x Difference amplifier or differential amplifier

A difference amplifier consists of an op amp where first input voltage v_1 is connected to v_- through an input resistance R_1 (current from v_1 to v_-), second input voltage v_2 is connected to v_+ through an input resistance R_2 (current from v_2 to v_+), v_+ is connected to ground through a ground resistance R_g (current from v_+ to ground), and v_o is fed back to v_- through a

feedback resistance R_f (current from v_- to v_o). Denote common-mode signal $v_{cm} = \frac{v_1 + v_2}{2}$ and differential signal $v_d = v_2 - v_1$.

$$v_o = -\frac{R_f}{R_1} v_1 + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1}\right) v_2$$

$$i_1 = \frac{1}{R_1} \left(v_1 - \frac{R_g}{R_2 + R_g} v_2\right)$$

$$i_2 = \frac{v_2}{R_2 + R_g}$$

Common-mode gain A_{cm} and differential gain A_d, defined by

$$v_o = v_{cm} A_{cm} + v_d A_d,$$

are

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2}{R_g} \frac{R_f}{R_1}\right)$$

$$A_d = \frac{1}{2} \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1}\right)$$

Common-mode rejection ratio (CMRR) is defined as

$$\text{CMRR} = 20 \log \left| \frac{A_d}{A_{cm}} \right|$$

To make $A_{cm} = 0$, we need

$$\frac{R_f}{R_1} = \frac{R_g}{R_2},$$

at which

$$A_d = \frac{R_f}{R_1} = \frac{R_g}{R_2}$$

Differential input resistance R_d is

$$R_d = R_1 + R_2$$

Thus, high differential gain hurts differential input resistance, which lower the voltage applied to the amplifier since source has internal resistance.

Proof. Use superposition. With v_1 enabled and v_2 disabled,

$$v_+ = v_- = 0$$

$$i_2 = i_g = 0$$

$$i_1 = \frac{v_1}{R_1} = i_f$$

$$v_o = -i_f R_f = -\frac{R_f}{R_1} v_1$$

With v_2 enabled and v_1 disabled,

$$i_2 = i_g = \frac{v_2}{R_2 + R_g}$$

$$v_+ = v_- = i_g R_g = \frac{R_g}{R_2 + R_g} v_2$$

$$i_1 = -\frac{v_-}{R_1} = -\frac{R_g}{R_1(R_2 + R_g)} v_2 = i_f$$

$$v_o = v_- - i_f R_f = \frac{R_g}{R_2 + R_g} v_2 + \frac{R_g R_f}{R_1(R_2 + R_g)} v_2 = \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1}\right) v_2$$

Sum:

$$v_o = -\frac{R_f}{R_1} v_1 + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1}\right) v_2$$

$$i_1 = \frac{v_1}{R_1} - \frac{R_g}{R_1(R_2 + R_g)} v_2 = \frac{1}{R_1} \left(v_1 - \frac{R_g}{R_2 + R_g} v_2\right)$$

$$i_2 = \frac{v_2}{R_2 + R_g}$$

$$v_1 = v_{cm} - \frac{v_d}{2}$$

$$v_2 = v_{cm} + \frac{v_d}{2}$$

$$\begin{aligned} v_o &= -\frac{R_f}{R_1} \left(v_{cm} - \frac{v_d}{2}\right) + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1}\right) \left(v_{cm} + \frac{v_d}{2}\right) \\ &= \frac{R_g}{R_2 + R_g} \left(1 - \left(\frac{R_2 + R_g}{R_g} - 1\right) \frac{R_f}{R_1}\right) v_{cm} + \frac{R_g}{R_2 + R_g} \left(1 + \left(\frac{R_2 + R_g}{R_g} + 1\right) \frac{R_f}{R_1}\right) \frac{v_d}{2} \\ &= \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2}{R_g} \frac{R_f}{R_1}\right) v_{cm} + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1}\right) \frac{v_d}{2} \end{aligned}$$

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2}{R_g} \frac{R_f}{R_1}\right)$$

$$A_d = \frac{1}{2} \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1}\right)$$

To make $A_{cm} = 0$, we need

$$1 = \frac{R_2}{R_g} \frac{R_f}{R_1}$$

$$\frac{R_f}{R_1} = \frac{R_g}{R_2}$$

at which

$$\begin{aligned} A_d &= \frac{1}{2} \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1}\right) \\ &= \frac{1}{2} \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_2}\right) \\ &= \frac{R_g}{R_2 + R_g} \frac{R_2 + R_g}{R_2} \\ &= \frac{R_g}{R_2} \end{aligned}$$

Differential input resistance:

$$i_d = \frac{v_+}{R_g} = \frac{v_2 - v_+}{R_2} = \frac{v_+ - v_2 + v_d}{R_1}$$

$$R_2 v_+ = R_g v_2 - R_g v_+$$

$$R_1 v_2 - R_1 v_+ = R_2 v_+ - R_2 v_2 + R_2 v_d$$

$$v_2 = \left(\frac{R_2 + R_g}{R_g} \right) v_+$$

$$(R_1 + R_2)(v_2 - v_+) = R_2 v_d$$

$$v_+ = \frac{R_g}{R_1 + R_2} v_d$$

$$i_d = \frac{1}{R_1 + R_2} v_d$$

$$R_d = \frac{v_d}{i_d} = R_1 + R_2$$

□

xi Instrumentation amplifier (INA)

An instrumentation amplifier consists of a difference amplifier (second stage) with outputs of two identical non-inverting amplifiers (first stage) as inputs, which behaves the same as the difference amplifier but with both common-mode and differential gains multiplied by the gain of the first stage. It can achieve higher differential gain than difference amplifier under same differential input resistance. However, common-mode signal is also amplified in the first stage, so CMRR of second stage must be good, i.e., the resistances must be precise.

xii Inverting configuration with general impedance

Consider an op amp where input voltage v_i is connected to v_- through an input impedance Z_i (current from v_i to v_-), v_+ is connected to ground, and v_o is fed back to v_- through a feedback impedance Z_f (current from v_- to v_o).

Laplace domain:

$$V_+ = V_- = 0$$

$$I_i = \frac{V_i}{Z_i} = I_f = -\frac{V_o}{Z_f}$$

$$V_o = -\frac{Z_f}{Z_i} V_i$$

Effect of finite open-loop gain:

$$V_+ = 0$$

$$V_o = -A V_-$$

$$I_i = \frac{V_i - V_-}{Z_i} = \frac{V_i + \frac{V_o}{A}}{Z_i} = I_f = -\frac{V_o + \frac{V_o}{A}}{Z_f}$$

$$G = \frac{V_o}{V_i} = \frac{-\frac{Z_f}{Z_i}}{1 + \frac{1 + \frac{Z_f}{Z_i}}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{1 - G_{\text{ideal}}}{A}}$$

$$\lim_{A \to \infty} G = -\frac{Z_f}{Z_i} = G_{\text{ideal}}$$

For single-pole model $A(s) = \frac{A_0}{1 + \frac{s}{\omega_b}}$ with $A_0 \gg 1 + \frac{Z_f}{Z_i}$ and $\omega_t = A_0\omega_b$:

$$\begin{aligned} G &= \frac{-\frac{Z_f}{Z_i}}{1 + \frac{1+\frac{Z_f}{Z_i}}{A}} \\ &= \frac{-\frac{Z_f}{Z_i}}{1 + \frac{1+\frac{Z_f}{Z_i}}{A_0} + \frac{s}{\frac{\omega_t}{1+\frac{Z_f}{Z_i}}}} \\ &\approx \frac{-\frac{Z_f}{Z_i}}{1 + \frac{s}{\frac{\omega_t}{1+\frac{Z_f}{Z_i}}}}, \end{aligned}$$

xiii Inverting integrator or Miller integrator

An inverting integrator consists of an op amp where input voltage v_I is connected to v_- through an input resistance R (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a feedback capacitance C (current from v_- to v_O).

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Time domain:

$$v_+ = v_- = 0$$

$$i_I = \frac{v_I}{R} = i_C = -C\frac{\mathrm{d}v_O}{\mathrm{d}t}$$

$$v_O = v_O(0^-) - \frac{1}{RC}\int_0^t v_I(\tau)\,\mathrm{d}\tau$$

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_C = -CsV_O + Cv_O(0^-)$$

$$V_O = \frac{1}{s}v_O(0^-) - \frac{1}{RCs}V_I$$

Transfer function: Infinite DC gain. Thus, any tiny DC component in the input signal will blow this out. $\frac{\pi}{2}$ phase shift.

$$H(s) = \frac{V_o}{V_i} = -\frac{1}{RCs}$$

Unity-gain angular frequency:

$$\frac{1}{RC}$$

xiv Inverting integrator or Miller integrator feasible configuration

An inverting integrator feasible configuration consists of an op amp where input voltage v_I is connected to v_- through an input resistance R (current from v_I to v_-), v_+ is connected to

ground, and v_O is fed back to v_- through a feedback capacitance C and feedback resistance R_f in parallel connection with the sum of their current be i_f (from v_- to v_O).

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_f = -\left(\frac{1}{R_f} + Cs\right)V_O + Cv_O(0^-) = -\frac{1 + R_f Cs}{R_f}V_O + Cv_O(0^-)$$

$$V_O = \frac{R_f C}{1 + R_f Cs}v_O(0^-) - \frac{\frac{R_f}{R}}{1 + R_f Cs}V_I$$

Transfer function: STC low-pass with cutoff angular frequency $\frac{1}{R_f C}$ and DC gain $\frac{R_f}{R}$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R_f}{R}}{1 + R_f Cs}$$

xv Inverting differentiator

An inverting differentiator consists of an op amp where input voltage v_I is connected to v_- through an input capacitor C (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a feedback resistor R (current from v_- to v_O).

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Time domain:

$$i_I = C\frac{\mathrm{d}v_I}{\mathrm{d}t} = i_f = \frac{-v_O}{R}$$

$$v_O = -RC\frac{\mathrm{d}v_I}{\mathrm{d}t}$$

Laplace domain:

$$I_I = CsV_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -RCsV_I + RCv_I(0^-)$$

Transfer function: Infinite gain at $\omega = \infty$. $-\frac{\pi}{2}$ phase shift.

$$H(s) = \frac{V_o}{V_i} = -RCs$$

Unity-gain angular frequency:

$$\frac{1}{RC}$$

xvi Inverting differentiator feasible configuration

An inverting differentiator feasible configuration consists of an op amp where input voltage v_I is connected to v_- through an input capacitor C and a input resistor R_I in series connection (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a feedback resistor R (current from v_- to v_O).

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Laplace domain:

$$I_I = \frac{1}{R_I + \frac{1}{Cs}} V_I - C v_I(0^-) = \frac{\frac{s}{R_I}}{s + \frac{1}{R_I C}} V_I - C v_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -\frac{\frac{R}{R_I} s}{s + \frac{1}{R_I C}} V_I + RC v_I(0^-)$$

Transfer function: STC high-pass with cutoff angular frequency $\frac{1}{R_I C}$ and gain $\frac{R}{R_I}$ at $\omega = \infty$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R}{R_I} s}{s + \frac{1}{R_I C}}$$

xvii Large-signal operation

- Output voltage saturation: A real op amp operates linearly over a limited range of output voltages and saturates if the output voltage reaches a saturation voltage, which is related to the positive and negative power supply voltages. An op amp with saturation voltage equal to its power supplies is said to offer rail-to-rail output.
- Output current limits: A real op amp's output current is limited to a specified maximum. This also imposes a output voltage limit equal to the maximum output current multiplied by the sum of output impedance and load impedance. For 741, its maximum output current is 20 mA.

xviii Slew rate (SR)

There is a specific maximum rate of change possible at the output voltage of a real op amp, called slew rate (SR),

$$\mathrm{SR} = \left| \frac{\mathrm{d}v_o}{\mathrm{d}t} \right| \Bigg|_{\max},$$

usually specified on the op-amp data sheet in units of V/μs.

To measure it, connect v_o to v_- and v_+ to a voltage source v_t and measure the rate of change of v_o before it reaches v_t. The op amp is then said to be slewing, and the output is slew-rete limited.

3 Band Theory (能帶理論) and Semiconductor

I Basics

i Electronic band structure or band structure

The electronic band structure (or simply band structure) of a solid describes the range of energy levels that electrons may or may not have within it.

ii (Energy) band (能帶)

Atomic orbitals from multiple atoms combine to form molecular orbitals, such as bonding and anti-bonding orbitals. When a vast number of atoms—typically on the order of 10^{20} or more—come together to form a solid, the energy differences between these molecular orbitals become extremely small. Consequently, a cluster of closely spaced energy levels can be treated as a continuous energy band.

iii Fermi level (費米能階)

The Fermi level of a solid-state body is the thermodynamic work required to add one electron to the body. For Cu, $E_F = 7.0$ eV.

iv Work function (功函數)

For a material with Fermi level E_F and energy of an electron infinitely far in vacuum E_{vac}, the work function is

$$W = E_{\text{vac}} - E_F.$$

Work function is related with the threshold frequency f_0 of photoelectric effect by

$$W = hf_0.$$

v Density or concentration

For a type of particles, the (number) density or concentration of it is defined as the number of it per unit volume in a material.

vi Number density of states of the allowed energy level per unit volume per unit energy interval

The number density of states of the allowed energy level per unit volume per unit energy interval as a function of energy level E is

$$N(E) = \frac{8\sqrt{2}\pi m_e^{3/2}}{h^3}\sqrt{E}$$

where m_e is the mass of an electron.

vii Occupancy probability

The occupancy probability, the probability that a given available state will be occupied by an electron, as a function of energy level E, is

$$P(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$$

where T is temperature in K and k is Boltzmann's constant 8.62×10^{-5} eV/K.

viii Number density of occupied states per unit volume per unit energy interval

The number density of occupied states per unit volume per unit energy interval $N_0(E)$ is

$$N_0(E) = N(E)P(E).$$

ix Finding Fermi energy

E_F can be found by solving

$$n = \int_0^{E_F} N_0(E)\,\mathrm{d}E\,.$$

The result is

$$E_F = \left(\frac{3}{16\sqrt{2}\pi}\right)^{2/3} \frac{h^2}{m_e} n^{2/3} \approx \frac{0.121h^2}{m_e} n^{2/3}.$$

x Ohmic contact

An ohmic contact is a non-rectifying electrical junction (接面), that is, a junction between two semiconductors or conductors that has a linear current–voltage (I–V) curve as with Ohm's law.

II Metals

i Bands

A partially filled band always exist, in which electrons can wander around and conduct current, called free electrons (or simply electrons).

ii Density of free electrons in partially filled band

Density of electrons in partially filled band n is the material's mass density multiplied by the material's number of valence electrons per atom multiplied by Avogadro's number divided by the material's molar mass.

iii Thermal effect

The change of Fermi level on temperature change is small and often negligible for metals.

iv Drude model of drift velocity and drift current

In Drude model, after a typical collision, each free electron completely lose its memory of its previous velocity and moving off in a random direction. In the average time τ between collisions, the average free electron will acquire a drift speed of

$$\mathbf{v}_d = -\frac{q\tau}{m}\mathbf{E}$$

where q is the elementary charge.

We define the free electron mobility μ_n as

$$\mu_n = \frac{q\tau}{m}$$

Suppose the free electron concentration is n. Then, the charge density is $-nq$ and the drift current density is

$$\mathbf{J}_d = -nq\mathbf{v}_d = nq\mu_n\mathbf{E} = \frac{nq^2\tau}{m}\mathbf{E}$$

Thus the resistivity ρ is

$$\rho = \frac{|\mathbf{E}|}{|\mathbf{J}|} = \frac{m}{nq^2\tau}$$

which is a constant, i.e., obeying Ohm's law.

III Semiconductors

i Bands

For semiconductors and insulators,

- **Conduction band (導帶)**: The lowest-energy band that is not filled, in which electrons, if exist, can wander around and conduct current, called free electrons (or simply electrons). The energy level of conduction band minimum is denoted E_C.
- **Valence band (價帶)**: The highest-energy band that is filled, in which electrons are constrained in covalent bonds and can't wander away freely or conduct current. The energy level of valence band maximum is denoted E_V.
- **Band gap (能隙)**: The gap between conduction band minimum and valence band maximum. The energy difference of the gap is called band-gap energy $E_g = E_C - E_V$. For silicon, $E_g = 1.12$ eV.

ii Insulator

A material with empty conduction band and high band-gap energy such that no transition of electron from valence band to conduction band occurs.

iii Semiconductor

A material whose elements of crystal structure have a valence of four in average and form a network of bonds as valence band and with low band-gap energy such that transition of electron from valence band to conduction band can occur under room temperature and that conductivity lies between that of conductors and insulators.

There are two kinds of semiconductors:

- single-element semiconductors, such as silicon (IV) and germanium (IV), and
- compound semiconductors, such as gallium arsenide GaAs (III-V), aluminium gallium arsenide AlGaAs (III-V), indium phosphide InP (III-V), aluminium gallium indium phosphide AlInGaP (III-V), zinc oxide ZnO_2 (II-VI), and titanium dioxide TiO_2 (II-VI).

Germanium was used in the fabrication of very early transistors (late 1940s, early 1950s). It was quickly supplanted by silicon, on which today's integrated-circuit technology is almost entirely based.

Compound semiconductors are useful in special electronic circuit applications as well as in applications that involve light, such as light-emitting diodes (LEDs).

iv Energy and effective mass (有效質量) of electrons

The energy of an electron in a semiconductor is not the simple square relation of that of a free electron and wave number but a complicated periodic function. However, near an edge of a band, it can be approximated with:

$$E(k) \approx E_0 + \frac{\hbar^2 k^2}{2m^*}$$

where E_0 is the energy of the edge, k is wave number, and m[8] is a constant called effective mass given by

$$m^* = \hbar^2 \left(\frac{\mathrm{d}^2 E(k))}{\mathrm{d}k^2} \right)^{-1}$$

where $\frac{\mathrm{d}^2 E(k)}{\mathrm{d}k^2}$ is taken at the edge. Effective mass is usually written as multiple of mass of electron.

v Effective mass of electrons at difference levels

- At conduction band minimum, $\frac{\mathrm{d}^2 E(k)}{\mathrm{d}k^2} > 0$ and $m^* > 0$. These electrons behave like free electrons but with mass m^*.
- At valence band maximum, $\frac{\mathrm{d}^2 E(k)}{\mathrm{d}k^2} < 0$ and $m^* < 0$. These electrons move to the opposite direction of a force acting on it.

vi (Electron) hole (（電子）電洞/空穴)

A hole h^+ is an absence of an electron in valence band.

Let the effective mass of electrons at valence band maximum be $-m^*$ and the concentration of holes be p. Consider a nearly-filled valence band, the current of it equals to the current of a filled valence band, which is always zero, subtracting the current that would exist due to the electrons that would fill the holes if there are not the holes, which is in the opposite direction of the force. Thus, the current is in the direction of the force and equals to that due to particles with charge q and mass m^* at concentration p.

Thus, we define hole as a quasiparticle with charge q and effective mass m^*, which is positive.

vii Carriers (載子) of semiconductors

Thus, there are two types of carriers in semiconductors mathematically:

- free electron at conduction band minimum with charge $-q$ and a positive effective mass m_n^*, and whose properties are denoted with subscript n, and
- holes at valence band maximum with charge q and a positive effective mass m_p^*, and whose properties are denoted with subscript p.

The concentration of free electrons is denoted n; the concentration of holes is denoted p.

viii Effective densities of states

Available electronic states effectively participate in carrier occupation near a band edge at a given temperature, denoted as N_C for conduction band minimum and N_V for valence band maximum. They are governed by

$$N_C = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$$

$$N_V = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

ix Carrier generation

When an electron transits from valence band to conduction band, a free electron in conduction band and a hole in valence band are generated, and an energy of band-gap energy E_g is consumed.

Since they are generated in pairs, an intrinsic (undoped) semiconductor has equal concentration of free electrons and holes.

x Carrier recombination (載子複合) and electroluminescence (電致發光/電激發光)

When an electron transits from conduction band to valence band, a free electron in conduction band and a hole in valence band are cancelled, and an energy of band-gap energy E_g is released in the form of a photon with frequency $f_g = \frac{E_g}{h}$, called electroluminescence.

Carrier recombination occurs naturally and at a rate proportional to the number of free electrons and holes.

Semiconductors with different band-gap energy can release light of different frequency. This effect is used in light emitting diode (LED) (發光二極體).

xi Exciton (激子)

A quasiparticle with no charge of a pair of free electron and hole in bounded state due to electrostatic force.

xii Photovoltaic effect (光生伏打效應/光伏效應)/Internal photoelectric effect (內光電效應)

Carrier generation caused by light.

Similar to the photoelectric effect, if the frequency of the incident light is greater than or equal to the material's threshold frequency $f_g = \frac{E_g}{h}$, electron-hole pairs are generated immediately; if the frequency of the incident light is less than this threshold, such pairs are never produced.

This effect is used in solar cell, photoresistor, and light-controlled relay.

xiii Thermal generation

Carrier generation due to thermal energy. The higher the temperature, the more thermal generations occur.

xiv Thermal equilibrium

Thermal equilibrium is the state where thermal generation rate equals to carrier recombination rate, at which

$$np = n_i^2$$

where n_i denotes the number of free electrons and holes in a unit volume at a given temperature given by

$$n_i = BT^{3/2}e^{-\frac{E_g}{2kT}}$$

where B is a material-dependent parameter that is $7.3 \times 10^{15}\ \text{cm}^{-3}\text{K}^{-\frac{3}{2}}$ for silicon, and

$$n = N_C \exp\left(-\frac{E_C - E_F}{kT}\right)$$

$$p = N_V \exp\left(-\frac{E_F - E_V}{kT}\right)$$

For intrinsic semiconductor at thermal equilibrium,

$$n = p = n_i$$

$$E_F = \frac{E_C + E_V}{2} + \frac{kT}{2}\ln\left(\frac{N_V}{N_C}\right) = \frac{E_C + E_V}{2} + \frac{3kT}{4}\ln\left(\frac{m_p^*}{m_n^*}\right)$$

At 0 K, no thermal generation occurs, so an intrinsic semiconductor has no carrier and behaves as an insulator at thermal equilibrium. At room temperature, the carrier concentration at thermal equilibrium is still far too small for intrinsic silicon to conduct appreciable current.

xv Doping (摻雜)

Doping is the intentional introduction of impurities into an intrinsic/undoped/pure semiconductor in order to change its properties. The doped material is referred to as an extrinsic/doped semiconductor. E_C and E_V are approximately unchanged for light doping.

xvi n-typed (n 型) semiconductors or n materials

Semiconductor with doping of elements that readily lose electrons, called donors, typically elements with valence of 5, such as phosphorus P and arsenic As, which introduces an increase of the concentration of free electrons, Fermi level, and conductivity. The introduced free electrons are neutralized by an equal amount of bound/bounded positive charge associated with the donor atoms even though they are often not shown in diagrams for simplicity.

If the concentration of donor atoms is N_D, which is usually far greater than n_i, the concentration of free electrons in the n-type semiconductor will be approximately

$$n_n = N_D$$

and, at thermal equilibrium,

$$p_n = \frac{n_i^2}{N_D}$$

Hence, free electrons are said to be the majority (charge) carriers, while holes are said to be the minority (charge) carriers. The minority holes are often not shown in diagrams for simplicity.

xvii p-typed (p 型) semiconductors or p materials

Semiconductor with doping of elements that readily get electrons, called acceptors, typically elements with valence of 3, such as boron B and aluminium Al, which introduces an increase of the concentration of holes and conductivity and decrease of Fermi level. The introduced holes are neutralized by an equal amount of bound/bounded negative charge associated with the acceptor atoms even though they are often not shown in diagrams for simplicity.

If the concentration of acceptor atoms is N_A, which is usually far greater than n_i, the concentration of holes in the p-type semiconductor will be approximately

$$p_p = N_A$$

and, at thermal equilibrium,

$$n_p = \frac{n_i^2}{N_A}$$

Hence, holes are said to be the majority (charge) carriers, while free electrons are said to be the minority (charge) carriers. The minority free electrons are often not shown in diagrams for simplicity.

IV Current flow in semiconductors

i Drude model of drift velocity and drift current

In Drude model, after a typical collision, each free electron or hole completely lose its memory of its previous velocity and moving off in a random direction. In the average time τ_n, τ_p between collisions, the average free electron and hole will acquire a drift speed of

$$\mathbf{v}_n = -\frac{q\tau_n}{m_n^*}\mathbf{E}$$

$$\mathbf{v}_p = \frac{q\tau_p}{m_p^*}\mathbf{E}$$

We define the free electron mobility μ_n and hole mobility μ_p as

$$\mu_n = \frac{q\tau_n}{m_n^*}$$

$$\mu_p = \frac{q\tau_p}{m_p^*}$$

For silicon, μ_n is 1350 $\mathrm{cm^2V^{-1}s^{-1}}$ and μ_p is 480 $\mathrm{cm^2V^{-1}s^{-1}}$.

The charge densities are $-nq$ and pq for free electrons and holes respectively, and the drift current densities are

$$\mathbf{J}_{S,n} = -nq\mathbf{v}_n = nq\mu_n\mathbf{E} = \frac{nq^2\tau_n}{m_n^*}\mathbf{E}$$

$$\mathbf{J}_{S,p} = pq\mathbf{v}_n = pq\mu_p\mathbf{E} = \frac{pq^2\tau_p}{m_p^*}\mathbf{E}$$

The total drift current density is

$$\mathbf{J}_S = \mathbf{J}_{S,n} + \mathbf{J}_{S,n} = q\left(n\mu_n + p\mu_p\right)\mathbf{E} = q^2\left(\frac{n\tau_n}{m_n^*} + \frac{p\tau_p}{m_p^*}\right)\mathbf{E}$$

Thus the resistivity ρ is

$$\rho = \frac{|\mathbf{E}|}{|\mathbf{J}_S|} = \frac{1}{q\left(n\mu_n + p\mu_p\right)} = \frac{1}{q^2\left(\frac{n\tau_n}{m_n^*} + \frac{p\tau_p}{m_p^*}\right)}$$

which is a constant, i.e., obeying Ohm's law.

ii Diffusion current

When the concentration of a type of charge carriers differ across a piece of semiconductor, the charge carriers will diffuse from he region of high concentration to the region of low concentration, which gives rise to a net flow of charge, called diffusion current.

The concentration of charge carriers of a piece of semiconductor under a state is called concentration profile.

The diffusion current densities of free electrons and holes are

$$\mathbf{J}_{D,n} = -qD_n\nabla n$$

$$\mathbf{J}_{D,p} = qD_p\nabla p$$

where ∇_n, ∇_p are concentration gradients, D_n, D_p are constants called diffusion constants or diffusivities of free electrons and holes respectively given by:

$$D_n = \mu_n V_T$$

$$D_p = \mu_p V_T$$

called Einstein relationship, where V_T is thermal voltage given by

$$V_T = \frac{kT}{q}$$

For $T = 300$ K, V_T is 25.9 mV. For intrinsic silicon, D_n is 35 cm^2/s and D_p is 12 cm^2/s.

V pn or p-n junction at equilibrium

i Physical structure

A p-n junction is a single semiconducting crystal with one end doped to form p material and the other end doped to form n material. The two types meet at a junction plane. External wire

connections are made to both ends through metal contacts. If the pn junction is used as diode (二極體), these constitute the two terminals, anode, indicated by +, for p-type end and cathode, indicated by -, for n-type end, of the diode. A one-dimensional coordinate is established with junction as 0 and towards n region as positive direction.

ii Abrupt junction

A pn junction where the doping concentration is made to change abruptly at the junction boundary is called abrupt junction.

A pn junction where the doping concentration is made to change gradually from one side of the junction to the other is called graded junction.

We will be studying abrupt pn junction unless otherwise specified.

iii (Carrier-)depletion/space-charge region/layer/zone and diffusion current

- Majority holes diffuse from the p region to the n region. The holes that diffuse across the junction into the n region quickly recombine with some of the majority free electrons, which results in the disappearance of some free electrons in the n region. Thus, some bound positive charge will no longer be neutralized by free electrons, and this charges are said to have been uncovered. Since recombination occurs close to the junction, there will be a region in the n region near the junction that is depleted of free electrons and contains uncovered bound positive charge.
- Majority free electrons diffuse from the n region to the p region. The free electrons that diffuse across the junction into the p region quickly recombine with some of the majority holes, which results in the disappearance of some holes in the p region. Thus, some bound negative charge will no longer be neutralized by holes, and this charges are said to have been uncovered. Since recombination occurs close to the junction, there will be a region in the p region near the junction that is depleted of holes and contains uncovered bound negative charge.

The region on both sides of the junction, with the n side of this region positively charged and depleted of free electrons and the p side negatively charged and depleted of holes, is called depletion region.

Due to the depletion, the carrier concentrations of holes and free electrons in the whole depletion region are both negligible.

Thus, the net charge density in the region is the bound charge density.

- In n-type part of the depletion region, the net charge density is N_D.
- In p-type part of the depletion region, the net charge density is $-N_A$.

The width of the region on the two sides are not always the same. Rather, to uncover the same amount of charge, the depletion layer will extend deeper into the more lightly doped material, that is, wider into the n material if $N_A > N_D$ and wider into the p material if $N_A < N_D$, where N_A and N_D are the doping concentrations of the p side and n side of the junction respectively.

The charges on both sides cause an electric field **E** to be established in the direction from n side to p side, which results in a potential difference across the region, with the n side at a positive voltage relative to the p side, called barrier voltage V_0.

Both diffusions of majority carriers form currents from the n side to the p side. However, the barrier voltage acts as an energy barrier that only majority carriers with energy no less than qV_0 can diffuse pass the depletion region, and the others will be pushed back by the electric field.

- For the majority holes from the p region, the pushed back holes form a drift current density $pq\mathbf{E}$, which is from the p side to the n side, and thus the net current density due to majority holes from the p region to the n region through the depletion region is

$$\mathbf{J}_{D,p} = q\left(D_p \nabla p + p\mathbf{E}\right)$$

- For the majority free electrons from the n region, the pushed back free electrons form a drift current density $nq\mathbf{E}$, which is from the p side to the n side, and thus the net current density due to majority free electrons from the n region to the p region through the depletion region is

$$\mathbf{J}_{D,n} = q\left(-D_n \nabla n + n\mathbf{E}\right)$$

Their sum is the net current density due to majority carriers from the p region to the n region through the depletion region

$$\mathbf{J}_D = \mathbf{J}_{D,p} + \mathbf{J}_{D,n} = q(D_p \nabla p - D_n \nabla n + (n+p)\mathbf{E}),$$

The net current due to majority carriers from the p region to the n region through the depletion region is called diffusion current I_D, which decreases as barrier voltage increases.

The regions of the p and n regions outside the depletion region are called bulk regions.

iv Drift current

- Some of the thermally generated minority holes in the n region move toward the junction and reach the edge of the depletion region. There, they are swept across depletion region into the p side by the electric field.
- Some of the thermally generated minority free electrons in the p region move toward the junction and reach the edge of the depletion region. There, they are swept across depletion region into the n side by the electric field.

Both currents are from the n side to the p side, and their sum is called the drift current I_S. Since the current is carried by thermally generated minority carriers, its value is dependent on temperature; however, it is independent of barrier voltage because the drift current is determined by the number of minority carriers that make it to the edge of the depletion region; any minority carriers that manage to get to the edge of the depletion region will be swept across.

v Current equilibrium with open-circuit terminals

Under open-circuit condition, no external current exists; thus

$$I_D = I_S$$

This equilibrium is stable and maintained by V_0. If for some reason, $I_D > I_S$, more bound charges will be uncovered on both sides of the junction, which increases V_0 and thus decreases I_D, which in turns shrinks the depletion region since some of the uncovered bound charges are then covered.

vi Voltage equilibrium with short-circuit terminals

At equilibrium, the barrier voltage is balanced out by the voltages across the two metal-semiconductor junctions. Thus, no voltage can be measured across the terminals, complying with the conservation of energy.

vii (Junction) built-in voltage

The barrier voltage V_0 with no external voltage applied is given by

$$V_0 = V_T \ln\left(\frac{N_A N_T}{n_i^2}\right),$$

called the junction built-in voltage, which is typically in the range of 0.6 V to 0.9 V for silicon at room temperature.

viii Carrier concentration under no external voltage

Under no external voltage, which is signified by a subscript 0,

- at the p material edge of the depletion and further into the p material, the majority holes have concentration $p_{p0} = N_A$, and the minority free electrons have concentration $n_{p0} = \frac{n_i^2}{N_A}$;
- at the n material edge of the depletion and further into the n material, the majority free electrons have concentration $n_{n0} = N_D$, and the minority holes have concentration $p_{n0} = \frac{n_i^2}{N_D}$.

ix Width of and charge stored in depletion region

Let the materials be cuboid, cross sectional area of the junction be A, width of depletion region in p side and n side be x_p and x_n respectively, the total charge stored in depletion region in p side and n side be $Q_- < 0$ and $Q_+ > 0$ respectively. Then,

$$\left|Q_+\right| = qAx_nN_D = \left|Q_-\right| = qAx_pN_A$$

The width of the depletion region $W = x_p + x_n$ is given by the Poisson's equation

$$W = \sqrt{\frac{2\varepsilon}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_0}$$

where ε is the electrical permittivity of the material, which for silicon is $11.7\varepsilon_0 = 11.7 \times 11.7 \times 8.85 \times 10^{-14}$ F/cm $= 1.04 \times 10^{-12}$ F/cm. Typical W ranges from 0.1 μm to 1 μm.

$$x_n = W\frac{N_A}{N_A + N_D}$$

$$x_p = W\frac{N_D}{N_A + N_D}$$

The charge stored on either side of the depletion region is

$$\begin{aligned} Q_j &= \left|Q_+\right| = \left|Q_-\right| \\ &= WAq\frac{N_A N_D}{N_A + N_D} \\ &= A\sqrt{2\varepsilon q\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_0} \end{aligned}$$

We denote

$$\alpha = A\sqrt{2\varepsilon q\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

$$Q_j = \alpha\sqrt{V_0}$$

x One-sided junction

For most practical devices, one side of the pn junction is much more heavily doped than the other, called one-sided junction, that is, either

$$N_A \gg N_D, \quad W \approx x_n \gg x_p$$

or

$$N_D \gg N_A, \quad W \approx x_p \gg x_n$$

VI pn junction with applied voltage

i Reverse bias and forward bias

An external voltage is applied to the pn junction, with the voltage of the p-type, i.e., anode relative to the n-type, i.e., cathode, being V.

The effective barrier voltage is

$$V_0 - V$$

- If $V < 0$, it is called reverse bias, and the reverse bias voltage is $V_R = -V$.
- If $V > 0$, it is called forward bias, and the forward bias voltage is $V_F = V$.

ii Width of and charge stored in depletion region

For $V < V_0$, the width of the depletion region $W = x_p + x_n$ is given by the Poisson's equation

$$W = \sqrt{\frac{2\varepsilon}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_0 - V)}$$

and the charge stored on either side of the depletion region is

$$Q_j = A\sqrt{2\varepsilon q\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_0 - V)} = \alpha\sqrt{V_0 - V}$$

iii Current-voltage relationship

- The concentration of minority holes in the n region at the edge of the depletion region will increase considerably to

  $$p_n(x_n) = p_{n0}e^{V/V_T}$$

  Excess concentration is defined as

  $$p_n(x_n) - p_{n0} = p_{n0}\left(e^{V/V_T} - 1\right)$$

  As the injected holes diffuse into the n material, some will recombine with the majority free electrons and disappear. The excess hole concentration decays exponentially with distance. The total hole concentration in the n material is given by

  $$p_n(x) = p_{n0} + \left(p_n(x_n) - p_{n0}\right)e^{-(x-x_n)/L_p} = p_{n0} + p_{n0}\left(e^{V/V_T} - 1\right)e^{-(x-x_n)/L_p}, \quad x \geq x_n,$$

  where L_p is a constant called diffusion length of holes in the n material.

  By differentiating $p_n(x)$, we obtain the diffusion current density:

  $$J_p(x) = q\frac{D_p}{L_p}\left(p_n(x_n) - p_{n0}\right)e^{-(x-x_n)/L_p} = q\frac{D_p}{L_p}p_{n0}\left(e^{V/V_T} - 1\right)e^{-(x-x_n)/L_p}, \quad x \geq x_n$$

  It does not change inside the depletion region, thus

  $$J_p(x) = q\frac{D_p}{L_p}\left(p_n(x_n) - p_{n0}\right) = q\frac{D_p}{L_p}p_{n0}\left(e^{V/V_T} - 1\right), \quad -x_p \leq x \leq x_n$$

  The current inside the depletion region is

  $$I_p = AJ_p(0) = Aq\frac{D_p}{L_p}\left(p_n(x_n) - p_{n0}\right) = Aq\frac{D_p}{L_p}p_{n0}\left(e^{V/V_T} - 1\right)$$

- The concentration of minority free electrons in the p region at the edge of the depletion region will increase considerably to

  $$n_p(-x_p) = n_{p0}e^{V/V_T}$$

  Excess concentration is defined as

  $$n_p(-x_p) - n_{p0} = n_{p0}\left(e^{V/V_T} - 1\right)$$

As the injected free electrons diffuse into the p material, some will recombine with the majority holes and disappear. The excess free electron concentration decays exponentially with distance. The total hole concentration in the p material is given by

$$n_p(-x) = n_{p0} + \left(n_p(-x_p) - n_{p0}\right) e^{-(x-x_p)/L_n} = n_{p0} + n_{p0}\left(e^{V/V_T} - 1\right) e^{-(x-x_p)/L_n}, \quad x \geq x_p,$$

where L_n is a constant called diffusion length of free electrons in the p material.

By differentiating $n_p(x)$, we obtain the diffusion current density:

$$J_n(-x) = q\frac{D_n}{L_n}\left(n_p(-x_p) - n_{p0}\right) e^{-(x-x_p)/L_n} = q\frac{D_n}{L_n} n_{p0}\left(e^{V/V_T} - 1\right) e^{-(x-x_p)/L_n}, \quad x \geq x_p$$

It does not change inside the depletion region, thus

$$J_n(-x) = q\frac{D_n}{L_n}\left(n_p(-x_p) - n_{p0}\right) = q\frac{D_n}{L_n} n_{p0}\left(e^{V/V_T} - 1\right), \quad x_p \geq x \geq -x_n$$

The current inside the depletion region is

$$I_n = AJ_n(0) = Aq\frac{D_n}{L_n}\left(n_p(-x_p) - n_{p0}\right) = Aq\frac{D_n}{L_n} n_{p0}\left(e^{V/V_T} - 1\right)$$

The total current inside the depletion region is

$$I = I_p + I_n = Aq\left(\frac{D_p}{L_p} p_{n0} + \frac{D_n}{L_n} n_{p0}\right)\left(e^{V/V_T} - 1\right)$$

Substituting $p_{n0} = \dfrac{n_i^2}{N_D}$ and $n_{p0} = \dfrac{n_i^2}{N_A}$ gives

$$I = Aqn_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right)\left(e^{V/V_T} - 1\right)$$

Taking $T = 0$, we obtain a constant current, which must be $-I_S$, thus

$$I_S = Aqn_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right)$$

$$I = I_S\left(e^{V/V_T} - 1\right)$$

called Shockley ideal diode equation.

iv Saturation current or scale current

We call I_S saturation current or scale current.

- It is proportional to cross-sectional area A.
- It is proportional to n_i^2, which strongly depends on T.
- It is the current in the reverse direction that the reverse bias situation saturates at.
- Typical values for I_S range from 10^{-18} A to 10^{-12} A.

v Reverse/junction breakdown

In reverse bias, as $-V$ reaches a breakdown voltage $V_{BR} \gg V_T$, the reverse current $-I$ will increase from I_S dramatically to a vary large value, and the reverse voltage across the junction remains very close to V_{BR}. This phenomenon is called junction breakdown. It is not a destructive phenomenon. That is, the pn junction can be repeatedly operated in the breakdown region without a permanent effect on its characteristics predicated on the assumption that the magnitude of the reverse-breakdown current is limited by the external circuit to a safe value that limits the power dissipated in the junction to a safe, allowable level.

There are two possible mechanisms for pn junction breakdown:

- Zener effect: It is the usual mechanism when $V_{BR} < 5$ V. The electric field in the depletion layer increases to the point of breaking covalent bonds and generating electron–hole pairs. The electrons generated in this way will be swept by the electric field into the n side and the holes into the p side. V_{BR} is often denoted V_Z in this case.
- avalanche effect: It is the usual mechanism when $V_{BR} > 8$ V. The minority carriers that cross the depletion region under the influence of the electric field gain sufficient kinetic energy to be able to break covalent bonds in atoms with which they collide. The carriers liberated by this process may have sufficiently high energy to be able to cause other carriers to be liberated in another ionizing collision.

When V_{BR} is between 5 V and 8 V, the breakdown mechanism can be either the zener or the avalanche effect or a combination of the two.

vi Conclusion

We conclude that

- in reverse bias without junction breakdown, I is negative and $-I$ is at most and often approximately I_S, which is very small, and thus the conductivity is very small.
- in forward bias, I is positive and often dominated by the exponential term, and thus the conductivity is large and increases rapidly as V increases.

VII Capacitive effects in pn junction

i Depletion or junction capacitance

Depletion capacitance predominates in reverse and zero bias and is from the charge stored on the depletion region given by

$$Q_j = A\sqrt{2\varepsilon q\left(\frac{1}{N_A}+\frac{1}{N_D}\right)(V_0+V_R)} = \alpha\sqrt{V_0+V_R}$$

In an ideal capacitor, the stored charge is linearly proportional to the applied voltage. In pn junction, however, the relationship is nonlinear. For small changes around a bias point, say

$V_R = V_Q$, we can define a depletion capacitance C_j as

$$C_j = \left.\frac{\mathrm{d}Q_j}{\mathrm{d}V_R}\right|_{V_R=V_Q}$$

Then,

$$C_j = \frac{\alpha}{2\sqrt{V_0 + V_R}}$$

The value of C_j at zero bias is

$$C_{j0} = \frac{\alpha}{2\sqrt{V_0}} = A\sqrt{\frac{\varepsilon q}{2}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\frac{1}{V_0}}$$

And

$$C_j = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}}$$

A more general form that also encompasses graded junction is

$$C_j = \frac{C_{j0}}{\left(1 + \frac{V_R}{V_0}\right)^m}$$

where m is a constant called grading coefficient, whose value ranges from 1/3 to 1/2 depending on the manner in which the concentration changes from the p to the n side.

ii Diffusion capacitance

Diffusion capacitance predominates only in forward bias. Consider a forward-biased pn junction. In steady state,

- Some holes injected from the p-side into the n-side become excess minority carriers in the n bulk region, which has charge:

$$\begin{aligned} Q_p &= Aq\int_{x_n}^{\infty} p_n(x) - p_{n0}\,\mathrm{d}x \\ &= Aq\int_{x_n}^{\infty} \left(p_n(x_n) - p_{n0}\right) e^{-(x-x_n)/L_p}\,\mathrm{d}x \\ &= Aq\left(p_n(x_n) - p_{n0}\right) L_p e^{-(x-x_n)/L_p}\Big|_{x_n}^{\infty} \\ &= \frac{L_p^2}{D_p} I_p \end{aligned}$$

  We define the time constant τ_p, called minority-carrier hole lifetime, as

$$\tau_p = \frac{L_p^2}{D_p}$$

$$Q_p = \tau_p I_p$$

- Some free electrons injected from the n-side into the p-side become excess minority carriers in the p bulk region, which has charge:

$$\begin{aligned}
Q_n &= Aq \int_{-\infty}^{-x_p} n_p(x) - n_{p0} \, \mathrm{d}x \\
Q_n &= Aq \int_{x_p}^{\infty} n_p(-x) - n_{p0} \, \mathrm{d} - x \\
&= Aq \int_{x_p}^{\infty} \left(n_p(-x_p) - n_{p0} \right) e^{-(x-x_p)/L_n} \, \mathrm{d} - x \\
&= Aq \left(n_p(-x_p) - n_{p0} \right) L_n e^{-(x-x_p)/L_n} \Big|_{x_p}^{\infty} \\
&= \frac{L_n^2}{D_n} I_n
\end{aligned}$$

  We define the time constant τ_n, called minority-carrier electron lifetime, as

$$\tau_n = \frac{L_n^2}{D_n}$$

$$Q_n = \tau_n I_n$$

The total excess minority-carrier charge is

$$Q = Q_p + Q_n = \tau_p I_p + \tau_n I_n$$

We define the (mean) transit time of the junction τ_T such that

$$Q = \tau_T I$$

For most practical one-sided junction, either

$$N_A \gg N_D, \quad I \approx I_p \gg I_n, \quad Q = Q_p \gg Q_n, \quad \tau_T \approx \tau_p$$

or

$$N_D \gg N_A, \quad I \approx I_n \gg I_p, \quad Q = Q_n \gg Q_p, \quad \tau_T \approx \tau_n$$

For small changes around a bias point, say $V = V_Q$, we can define an incremental diffusion capacitance C_d as

$$C_d = \left. \frac{\mathrm{d}Q}{\mathrm{d}V} \right|_{V=V_Q}$$

Using the approximation $e^{V/V_T} \gg 1$, $I = I_S \left(e^{V/V_T} - 1 \right) \approx I_S e^{V/V_T}$, we have

$$C_d = \frac{\tau_T}{V_T} I$$

C_d is directly proportional to the forward current I and thus is negligibly small when the diode is reverse biased. Also, to keep C_d small, the transit time τ_T must be made small, which is important for a pn junction intended for high-speed or high-frequency operation.